

DATA ANALYST PORTFOLIO

AI Evaluation of E-commerce Chatbot QA Responses for Prompt Robustness

Prepared by:

Patrick Adu Osei

IBM Data Analyst Professional Certificate Holder

Tools & Technologies:

Python (Pandas, Numpy) | Matplotlib/Seaborn| Jupyter Notebook | GitHub

Date: June 2026

DATA ANALYST PORTFOLIO

1. Summary

This project evaluates the robustness of AI-generated e-commerce chatbot responses under noisy, informal, and real-world customer language. Using a structured scoring framework, the evaluation measures whether the chatbot remains useful when prompts are typo-heavy, rude, ambiguous, incomplete, or multi-intent rather than neatly phrased.

The chart outputs show that the chatbot remains broadly stable across prompt types, with near-zero low-quality rates, very similar average scores between edge cases and standard prompts, and no strong evidence of catastrophic degradation under difficult prompts. However, the error taxonomy indicates that the dominant weakness is not factual failure but response optimization—especially generic phrasing, with secondary issues in verbosity and occasional omission of helpful steps.

2. Business / Analytical Question

The business and analytical question the evaluation is addressing are:

- How well do AI responses handle noisy, real-world customer language?
- Which conversational patterns lead to poor relevance or misunderstanding?
- Are customer intents correctly reflected in the AI response?
- Do rude, incomplete, or colloquial user prompts increase failure rates?

3. Dataset

Dataset used: `ecommerce_chatbot_qa.csv`. The dataset contains conversation-style chatbot interactions from which user prompts and assistant responses were parsed into evaluation-ready fields.

4. Tools Used

- Python (pandas, numpy) for parsing, scoring, and aggregation
- Jupyter Notebook for workflow execution and manual review
- Matplotlib and seaborn for chart generation
- CSV and PNG outputs for scored results, edge-case examples, and saved visuals

DATA ANALYST PORTFOLIO

5. Evaluation Process (What I Did)

1. Loaded the chatbot QA dataset and extracted user prompts and assistant responses from the conversation structure.
2. Built an evaluation data frame with robustness-focused dimensions: relevance, intent reflection, robustness to noisy prompts, task guidance, clarity, and business usefulness.
3. Tagged prompts into behavioural categories such as polite, rude/emotional, incomplete, ambiguous, multi-intent, and typo-heavy.
4. Applied semi-automated scoring and error tagging, then prepared results for focused review and aggregation.
5. Generated charts to compare performance by prompt type, low-quality rates, error taxonomy, edge-case vs standard prompts, and concrete edge-case examples.

6. Key Visualizations

1. Response Quality by Prompt Type compares the average overall score across behavioural prompt categories. The visible distribution suggests prompt-type performance is closely clustered rather than sharply separated, which points to stable robustness across messy customer language patterns.
2. Low-Quality Response Rate by Prompt Type shows prompt-type-level failure rates. The chart indicates that low-quality rates are effectively near zero across ambiguous, incomplete, multi-intent, polite, rude/emotional, and typo-heavy prompts, suggesting that the chatbot rarely drops below the defined low-quality threshold even under difficult prompt conditions.
3. Error Taxonomy Distribution shows the dominant failure patterns. The chart clearly indicates that overly generic responses dominate the error profile by a large margin, followed by missing key steps, verbosity, and weak prompt handling.
4. Average Score: Edge Cases vs Standard Prompts compares mean performance for flagged edge cases against standard prompts. The chart shows very similar averages, with edge cases appearing slightly higher or effectively equivalent, reinforcing the conclusion that difficult prompts do not significantly degrade baseline quality.
5. Edge-Case Examples Table provides concrete qualitative evidence. The visible examples show typo-heavy, rude/emotional, and multi-intent prompts receiving relevant but generic responses, with recurring error labels such as `overly_generic_response` and `occasional_too_verbose_or_inefficient`.

DATA ANALYST PORTFOLIO

7. Key Insights

1. The chatbot demonstrates strong robustness to noisy, messy, and emotionally charged language, as reflected in the near-zero low-quality rates across all prompt types.
2. There is no strong evidence that rude, incomplete, ambiguous, multi-intent, or typo-heavy prompts materially increase outright failure rates.
3. The dominant weakness is overly generic wording rather than factual or intent-level breakdown: the model usually stays relevant, but its answers are often not sufficiently specific or tailored.
4. Verbosity and missing key steps appear as secondary issues, indicating that some responses could be shorter, clearer, or more operationally complete.
5. Edge-case prompts are handled safely and consistently, but they still reveal shallow optimization: the model preserves tone and task alignment but often defaults to boilerplate-style guidance.

8. Business Implications and Recommendations (Why This Project Matters)

These findings suggest that the chatbot is broadly reliable for customer-support deployment in e-commerce settings because it remains relevant and stable even under difficult customer language patterns. That lowers the risk of obvious support failures in production.

However, the dominance of overly generic responses creates an important business risk: even when the system is technically correct, users may perceive it as repetitive, unhelpful, or less intelligent than expected. In e-commerce, this can reduce customer trust, weaken issue-resolution efficiency, and create friction in high-value flows such as cart actions, delivery queries, and payment-related support.

- Prioritize response specificity and context-aware wording rather than focusing only on raw correctness.
- Improve task guidance so that responses include clearer, more actionable operational steps when customers are trying to complete a purchase or resolve a support issue.
- Reduce boilerplate and unnecessary verbosity, especially in responses triggered by rude or typo-heavy prompts where users often want efficient help rather than generic reassurance.
- Preserve the current strong robustness to difficult prompts while improving usefulness and customer confidence through more tailored content.

DATA ANALYST PORTFOLIO

9. Challenges and How to Solve Them

- Challenge: The dataset is conversation-based and messy, which made parsing more difficult than working with flat tabular data.
Solution: Extract prompt / response pairs explicitly before evaluation.
- Challenge: Behavioural prompt tags can overlap (for example, a prompt can be both rude and typo-heavy).
Solution: Use a deterministic tagging rule for the current project, with the option to expand to multi-label tagging in future work.
- Challenge: Semi-automated scoring and rule-based error tagging can miss nuance.
Solution: Keep rules as a first pass, then validate key segments manually.
- Challenge: The strongest chart finding is not dramatic failure but the absence of failure across prompt types, which can be harder to narrate.
Solution: Frame robustness itself as a positive result, while using the error taxonomy to identify optimization priorities.

10. What I would Improve Next

To make the project even stronger in an evaluation setting, I would:

- Expand the manual review layer on a targeted subset of edge cases to create richer qualitative examples.
- Add inter-rater checks or re-scoring on a subset to measure evaluator consistency.
- Refine the taxonomy for generic responses into subtypes such as boilerplate reassurance vs generic instructions.
- Compare handcrafted prompts with naturally noisy prompts to quantify the robustness gap more precisely.
- Integrate this dataset with the Bitext and preference-comparison datasets to create a broader AI evaluation capstone covering both clean and messy prompt environments.

11. Files and Links

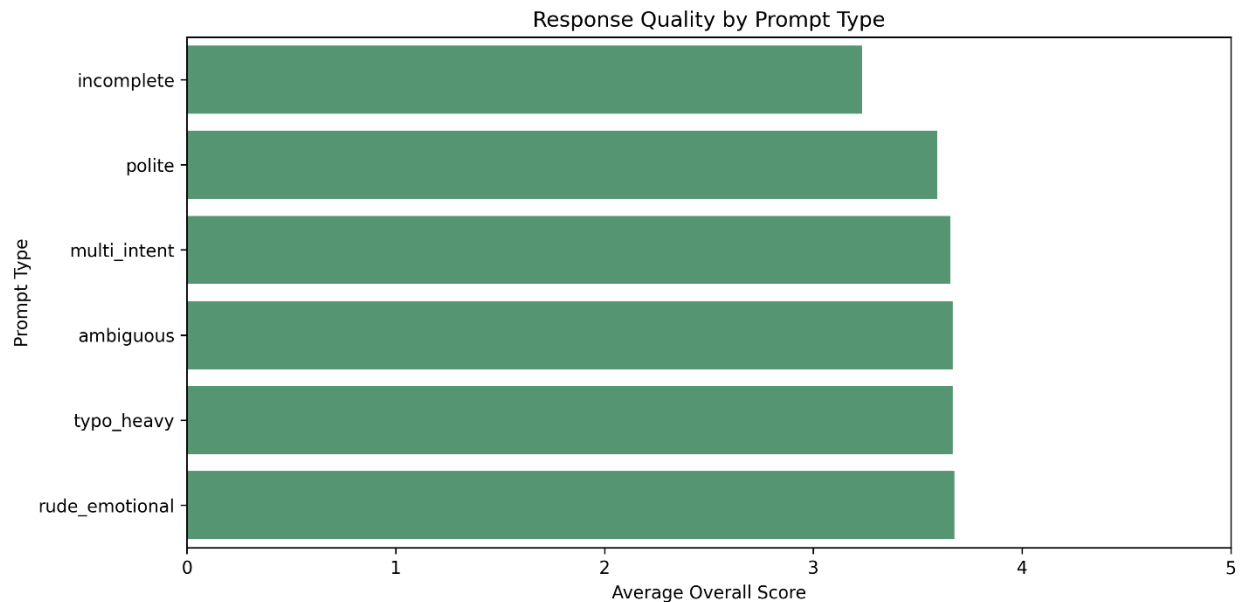
- Notebook: ecommerce chatbot QA evaluation notebook (.ipynb)
- Visual outputs: chatbot_qa_response_quality_by_prompt_type.png, chatbot_qa_error_taxonomy_distribution.png,

DATA ANALYST PORTFOLIO

chatbot_qa_low_quality_by_prompt_type.png,
chatbot_qa_edge_vs_normal_scores.png,
chatbot_qa_edge_case_examples_table.png

- Tabular outputs: chatbot_qa_edge_case_examples.csv,
chatbot_qa_scored_results.csv

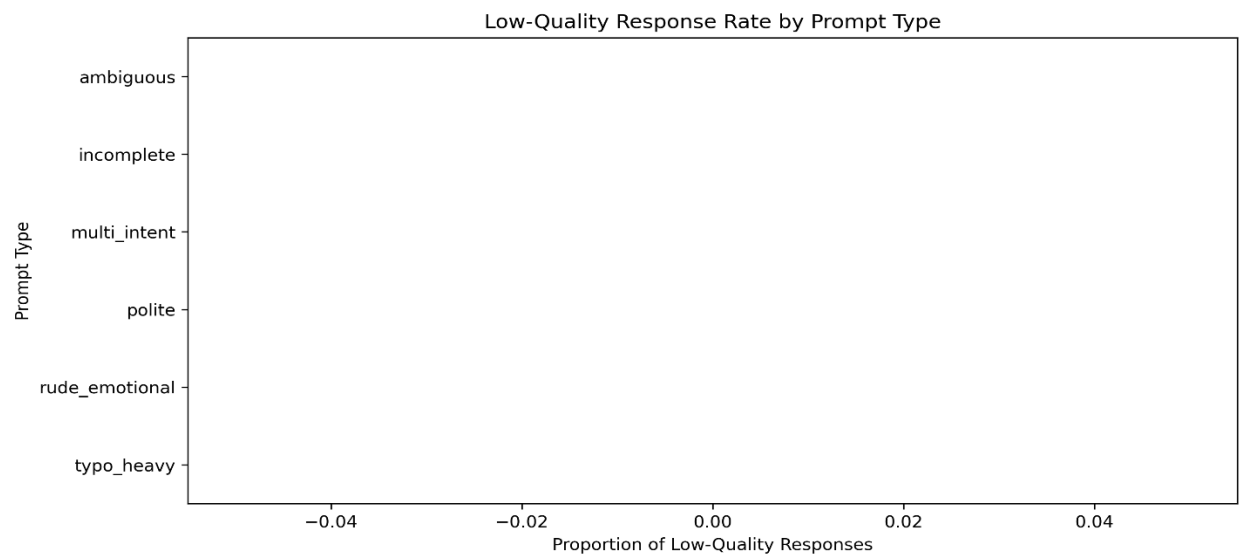
Chart 1: Response Quality by Prompt Type



- **Insight:** Performance is stable across behavioural prompt categories, suggesting that the chatbot handles noisy language without major quality collapse.

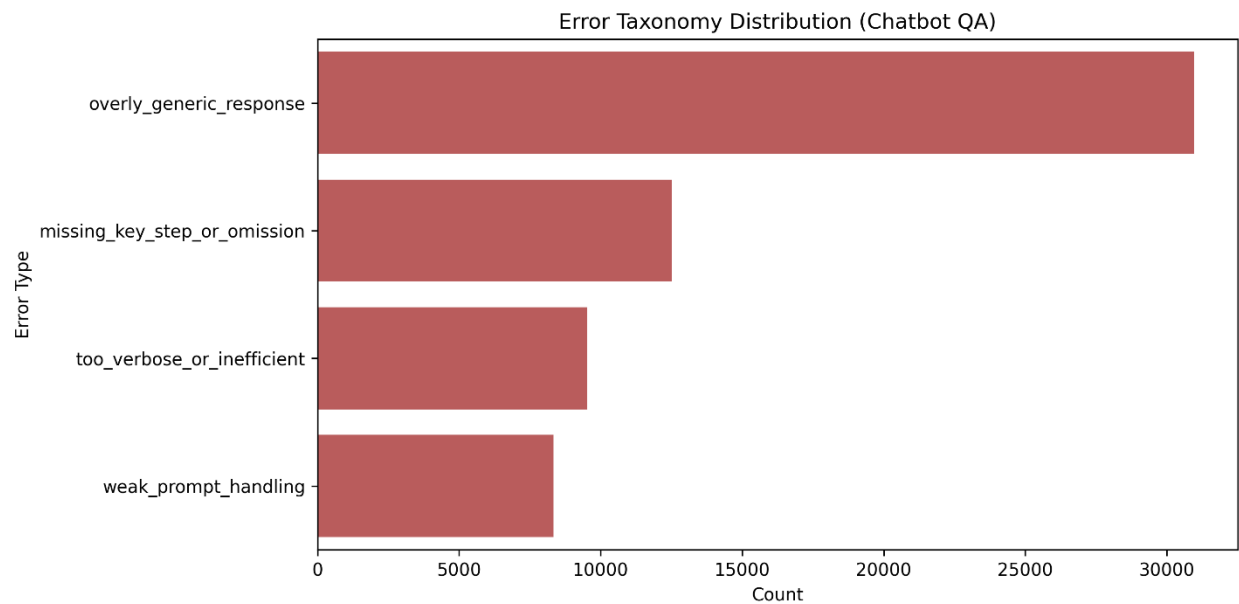
Chart 2: Low Quality Response Rate by Prompt Type

DATA ANALYST PORTFOLIO



- **Insight:** : Low-quality rates are effectively near zero across all prompt types, indicating strong baseline robustness.

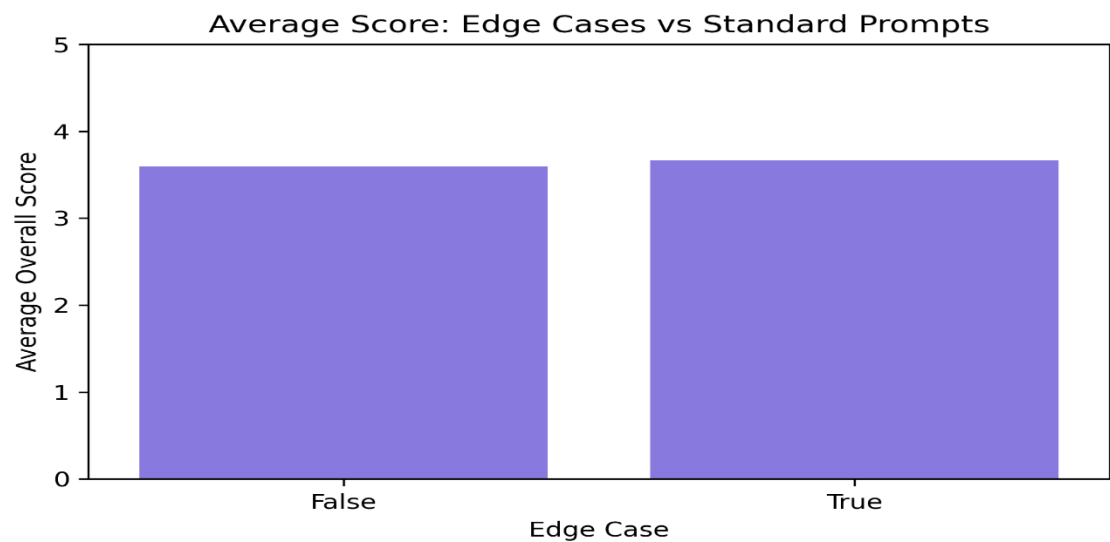
Chart 3: Error Taxonomy Distribution (Chatbot QA)



Insight: Generic responses dominate the error profile, which means optimisation should focus on specificity and tailoring rather than correcting widespread factual failures.

DATA ANALYST PORTFOLIO

Chart 4: Edge Cases vs Standard Prompts



- **Insight:** Edge cases perform roughly on par with standard prompts, supporting the conclusion that the model is robust to difficult customer language.

Chart 5: Edge Cases Examples

prompt	prompt_type	error_types	decision_summary	improvement_recommendation
I got to add an item to the cart.	type_heavy	overly_generic_response		
me add fucking products to the basket can hupel me	rude_emotional	overly_generic_response		
I have to add products to the basket I ned help	type_heavy	overly_generic_response; too_verbose_or_inefficient		
like to add products to the cart could I get some help	type_heavy	overly_generic_response		
need to add an item to the cart , where do I do it?	multi_intent	overly_generic_response		
fucking product to the basket, can I get some help ?	rude_emotional	overly_generic_response		
have to add items to the basket, where could I do it?	multi_intent	overly_generic_response; too_verbose_or_inefficient		
to add a fucking item to the basket , where to do it?	rude_emotional	overly_generic_response		
to add a fucking item of the cart can i get some help	rude_emotional	overly_generic_response		
t to add fucking items to the basket, how can I do it?	rude_emotional	overly_generic_response		

Insight: Difficult prompts are handled safely and politely, but often with generic or repetitive guidance rather than highly contextual support.